



Warmed Up

Lets Warm Up 



Easy

General Skills

picoCTF 2019

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Description

If I told you a word started with 0x70 in hexadecimal, what would it start with in ASCII?

Hints 

1

Submit your answer in our flag format. For example, if your answer was 'hello', you would submit 'picoCTF{hello}' as the flag.

148,955 users solved



75%

Liked



 picoCTF{FLAG}

Submit
Flag

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Challenge: Lets Warm Up

Category: General Skills

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I. Objective

The objective of this task was to convert a given **hexadecimal (base 16)** value into its corresponding **ASCII character** and submit the result in the required picoCTF flag format.

II. Background

Computers do not understand letters directly. Instead, they store and process **numbers**.

To make text readable, computers use a standard mapping system called **ASCII**, where each number represents a specific character.

Hexadecimal (base 16) is commonly used in computing because it is a compact way to represent binary data. Converting hexadecimal values to ASCII characters is a common skill in cybersecurity and reverse engineering.

III. Tool Used

- **Online ASCII / Hex Converter Website**
Used to automatically convert between hexadecimal, decimal, binary, base64, and ASCII formats.

IV. Methodology

1. Read the challenge description, which stated:

If a word started with 0x70 in hexadecimal, what would it start with in ASCII?

2. Opened an online converter capable of converting hexadecimal to ASCII.
3. Entered the hexadecimal value:

0x70

4. Observed the ASCII output provided by the tool.
5. The converter showed that 0x70 corresponds to the ASCII character:

p

6. Placed the result inside the picoCTF flag format.



V. Result

picoCTF{p}

Number delimiter

Space



☐ 0x/0b prefix

ASCII text

p

Hex (bytes)

70

Binary (bytes)

01110000

Decimal (bytes)

112

Base64

cA==

Length (bytes)



VI. Explanation

- The prefix 0x indicates that the number is written in **hexadecimal (base 16)**.
- The hex value 70 is first converted to decimal:
 $7 \times 16 + 0 \times 1 = 112$
- In the ASCII table, decimal value **112** corresponds to the lowercase letter:
p
- Therefore, a word that starts with 0x70 in hexadecimal starts with “**p**” in ASCII.
This process shows how computers convert numeric values into readable characters.

VII. Conclusion

This challenge demonstrates the relationship between **hexadecimal values and ASCII characters**. Understanding how to convert between number bases and text encodings is essential in cybersecurity, especially when analyzing memory, binaries, or encoded data. The correct ASCII interpretation of 0x70 leads directly to the flag.

— **The Analyst: Hyposelenia**